LIVE CLASS

(Placement & Competitive Exams)

TOPIC - DISTANCE & DIRECTION SENSE

Preassessment Question

Ankush went to the market to visit eight shops – Shop-1, Shop-2, Shop-3, Shop-4, Shop-5, Shop-6, Shop-7 and Shop-8, but not necessarily in the same order. He starts from Shop-1 and walks towards Shop-3 which is at distance of 24 m towards east. He then turns left and walks 16 m to reach Shop-5. Then he walks 57 m southwest to visit Shop-2, which is towards the south of Shop-1, crossing the path which goes through the midpoint of the line joining Shop-1 and Shop-3. After visiting Shop-2, he walked 40 m in east direction and reached Shop-7. He then walked 50 m left to reach Shop-8 and then walked 16 m right to reach Shop-4 and to reach the last shop he walked 35 m towards right.

Question: - What is the shortest distance between Shop-3 and Shop 6?

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Concept Building(From Basic)

1.Rohit walked 25 m towards south. Then he turned to his left and walked 20 m. He then turned to his left and walked 25 m. He again turned to his right and walked 15 m. At what distance is he from the starting point and in which direction?

A. 35m East B 35m North C.30m West D. 45m East

2. Sundar runs 20 m towards East and turns to right and runs 10 m. Then he turns to the right and runs 9 m. Again he turns to right and runs 5 m. After this he turns to left and runs 12 m and finally he turns to right and 6 m. Now to which direction is Sundar facing?

A. East

B. West

C. North

D. South

3.Golu started from his house towards North. After covering a distance of 8 km. he turned towards left and covered a distance of 6 km. What is the shortest distance now from his house?

A. 10 km.

B. 16 km.

C. 14 km

D. 2 km.

4.Ravi left home and cycled 10 km towards South, then turned right and cycled 5 km and then again turned right and cycled 10 km. After this he turned left and cycled 10 km. How many kilometers will he have to cycle to reach his home straight?

A. 10 km

B. 15 km

C. 20 km

D. 25 km

Directions (5-7): Study the information given below carefully and answer the questions that follow:
A, B, C, D, E, F, G, H and I are nine houses. C is 2 km east of B. A is 1 km north of B and H is 2 km south of A. G is 1 km west of H while D is 3 km east of G and F is 2 km north of G. I is situated just in middle of B and C while E is just in middle of H and D.

Q5. Distance between E and G is
(a) 1 km (b) 1.5 km (c) 2 km (d) 5 km (e) None of the above

Q6. Distance between E and I is
(a) 1 km (b) 2 km (c) 3 km (d) 4 km (e) None of the above

Q7. Distance between A and F is
(a) 1 km (b) 1.41 km (c) 2 km (d) 3 km (e) None of the above

Directions (8-9): Study the information given below carefully and answer the questions that follow:
On a playing ground, Dinesh, Kunal, Nitin, Atul and Prashant are standing as described below.
(i) Kunal is 40 metres to the right of Atul.
(ii) Dinesh is 60 metres to the south of Kunal.
(iii) Nitin is 25 metres to the west of Atul

(iv) Prashant is 90 metres to the north of Dinesh.

Q8. Who is to the north-east of the person who is to the left of Kunal?
(a) Dinesh (b) Nitin (c) Atul (d) Either Nitin or Dinesh (e) None of the above

Q9. If a boy walks from Nitin, meets Atul followed by Kunal, Dinesh and then Prashant, how many metres has he walked if he has travelled the straight distance all through?
(a) 155 metres (b) 185 metres (c) 215 metres (d) 245 metres (e) None of the above

SET 02

Q 1.A person is walking towards west for 25m later he turns left and walks 30 meters. Again he takes left and walks 50 meters. Later he takes left and walks 38 meters. Finally he walks 21 meters to his left. How far is he from the starting point and in which direction?

(1) 20 – SW (2) 80 – NE
(3) 4√5- NE (4) 5 – SW (5) None of the above

Learn the following table:

Important table

					Shortcut to learn the table
45^0		=	45^0	-	45× 1
45^0	$+45^0$	=	90^0	-	45× 2
90^0	$+45^0$	=	135^0	-	45× 3
135^0	$+45^0$	=	180^0	-	45× 4
180^0	$+45^0$	=	225^0	-	45× 5
225^0	$+45^0$	=	270^0	-	45× 6
270^0	$+45^0$	=	315^0	-	45× 7
315^0	$+45^0$	=	360^0	-	45× 8

Q 2.Shiva is facing N, he turns 180* and 45* in clockwise direction and then 270* in anti-clockwise direction. Finally again turns 180*. What is this facing direction now?

(1) SE (2) SW (3) South (4) North east (5) None of the above

3. P started from his house towards west. After walking a distance of 25 m. He turned to the right and walked 10 m. He then again turned to the right and walked 15 m. After this he is to turn right at 135° and to cover 30 m. In which direction should he is going?

A. West

B. South

C. South-West

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D. South-East

Q 4.A lady runs 12 km towards north. Then 6 km towards south and 8 km towards east. How for is she from her starting point and in which direction?

(1) NE – 20 km (2) NE – 15 km (3) NW – 10 km (4) NE – 10 km (5) None of the above

Q5.John's house is situated in North direction from his uncle's office at the distance of 100m.His uncle's house is situated in west from his uncle's office at the distance of 200m.Kabir is the friend of John, Kabir lives at 100m far in the east direction from John's house .Kabir's Office is in south direction from his house at the distance of 100m.How far is Uncle's house from Kabir's office?
(a) 200m (b) 300m (c) 400m (d)500m (e) none of these

Q6.A school driver drives school bus in north direction for a distance of 2km. Then he turns the bus left side and drives 5km straight. Then he turns bus in left side twice for a distance of 8km and 5km respectively. At last once more he took left turn and go for a distance of 1 km. How far is he from the starting point and in which direction he should drive for reaching starting point?
(a) 3km, North direction (b) 7km, West direction (c) 6km, South direction
(d) 6km, West direction (e) 5km, North direction

Q7.A man travels 2 km to the north and turns east and travels 10 km and again he turns north and travels 3 km and again turns to east and travels 2 km. How far is he from the starting point?
(a) 10km (b) 13km (c) 15km (d) 17km (e) none of these

Q8.Rajesh walked 8 km towards south-west direction. He turned to east direction and walked 20 km. Then he turned to north-east direction and walked 8km and after to west direction for 6km.Then he turned to north east direction and walked 2 km straight after he turned west direction and walked 4km.Once more he turned south-west and walked 2km.How far is he from the starting point?
(a)12km (b)10km (c)8km (d)6km (e)none of these

Directions (9-11): Study the following information carefully and answer the questions given below.
A poll was held in a particular constituency last month. Rakesh, who is a voter of that constituency, walked 40 metres towards east from his home and took a right turn and walked 20 metres. He again took a right turn and walked 30 metres. Finally he took a right turn and reached the polling station, which was 10 metres away from the previous point.

Q9.In which direction was the polling station located from the starting point?
(a)South (b) Southeast (c) Northeast (d) Northwest (e) East

Q10.What was the total distance covered by Rakesh to reach the polling station?
(a) 100m (b) 40m (c) 80m (d) 60m (e) 50m

Q11.In which direction was the polling station located from its first turning point?

(a)South-west (b) north-west (c) North-east (d) south-east (e) None of these

Q12.Ashok started walking towards South. After walking 50 m he took a right turn and walked 30 m. He then took a right turn and walked 100 m. He again took a right turn and walked 30 m and stopped. How far and in which direction was he from the starting point?
(a) 50 m South (b) 150 m North (c) 180 m East (d) 50 m North (e) None of these

SET 03:(Competitive Exams- Important)

SET-01

Directions : Study the information carefully and answer the questions given below.
Aman starts walking towards north direction from point A. After walking 6km he reached point B then turns towards his right and walks 9km to reach point C. Then he turns towards his right and walks 6km to reach point D.
Ronny starts walking from point E, which is mid-point of AB. He walks 5km towards east direction and reached point F. From there he turned towards his right and walks 7km and reached point G. From point G he turned towards his left and walk 4km to reach point H.

Q1. What is the distance between D and H?
(a) 4km
(b) 3km
(c) 5km
(d) 2km
(e) 1km.

Q2. In which direction point G with respect to A?
(a) East
(b) Northwest
(c) West
(d) Southeast
(e) None of these

Q3. What is the shortest distance between C and F?
(a) 1km
(b) 3km
(c) 5km
(d) 6km
(e) 1km

SET-02

Directions: Study the information carefully and answer the questions given below.
Point P is 10m west of point Q. Point X is 8m east of point M. Point Z is 10m north of

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point M. Point X is 5m south of point Q. Point B is 3m east of point Z.

Q4. What is the shortest distance between point P to X?
(a) 8m
(b) 5√5m
(c) 10m
(d) √13m
(e) None of these

Q5. In which direction point M with respect to point B?
(a) Southeast
(b) Northeast
(c) Southwest
(d) Northwest
(e) None of these

SET-03

Virendra's house is 30 m to the left of the Shani Dev Temple, which is 40 m to the north of the Church. One day, Virendra takes the road from his home which is parallel to the temple-Church road and walks for 50 m before turning to his right. He further walks for 20 m and reaches the Gym.
What is difference between the straight distance between Virendra's house to Church and Shani Dev Temple to Gym? (Root 2 = 1.41)

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SET-04

Anitha and Kathir are two friends studying in the XYZ Arts College. Anitha starts walking from her house in north direction. After walking 120m, she reached the Axis Bank. Then she turns and walks 40m in north-east direction. Then she turns 90 degree clock wise direction and walks 30m to reach Kathir's house which is in the east of the Axis Bank. Then from Kathir's house both of them walk 130m in south-east direction to reach their college which is east of Anitha's house.

What is the shortest distance between Anitha's House and XYZ Arts College?

What is the shortest distance between Anitha's House and Kathir's House?

SET-05

Ankush went to the market to visit eight shops – Shop-1, Shop-2, Shop-3, Shop-4, Shop-5, Shop-6, Shop-7 and Shop-8, but not necessarily in the same order. He starts from Shop-1 and walks towards Shop-3 which is at distance of 24 m towards east. He then turns left and walks 16 m to reach Shop-5. Then he walks 57 m southwest to visit Shop-2, which is towards the south of Shop-1, crossing the path which goes through the midpoint of the line joining Shop-1 and Shop-3. After visiting Shop-2, he walked 40 m in east direction and reached Shop-7. He then walked 50 m left to reach Shop-8 and then walked 16 m right to reach Shop-4 and to reach the last shop he walked 35 m towards right.

What is the direction of Shop-8 with respect to Shop-5?

A. Southwest
B. Northeast
C. East
D. West
E. Southeast

What is the direction of Shop-1 with respect to Shop-2 and the shortest distance between them?

A. Northeast, $7\sqrt{10}$ m
B. North, 35 m
C. Northwest, $7\sqrt{10}$ m
D. Northeast, 40 m
E. North, 39 m

What is the shortest distance between Shop-3 and Shop 6?

A. $6\sqrt{87}$ m
B. $12\sqrt{10}$ m
C. $8\sqrt{24}$ m
D. $4\sqrt{77}$ m
E. $4\sqrt{89}$ m

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